MS-ML Studio: methodological core for MS preprocessing, cross-validated feature extraction, and MLP classification

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Abstract

MS-ML Studio is an open-source MATLAB application that unifies mass-spectrometry (MS) preprocessing, three complementary feature extractors with embedded K-fold cross-validation, and multilayer perceptron (MLP) classification. This note summarizes the **algorithmic pipeline** implemented in the MS-ML Library and GUI: normalization, reshaping, optional Icoshift alignment, fuzzy moving-average peak detection, correlation- and probability-based peak selection, Watch Points, Feature Resonance with genetic-algorithm (GA) optimization, and MLP training. An illustrative intact-cell MALDI-TOF *Aeromonas* case study [1, 2] anchors partitions and metrics only; scope remains proof-of-concept software validation, not clinical benchmarking.

Keywords: mass spectrometry, feature extraction, genetic algorithm, MALDI-TOF, machine learning, open-source software

1. Introduction

MS workflows recur through import, preprocessing, feature extraction, and classification [3, 2, 4]. MS-ML Studio targets users who need these stages in one graphical session without scripting. The present text **focuses on**

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methods; narrative introduction, extended related work, and discussion appear in the full article under `latex/articlelsevier/`.

2. Materials and methods

2.1. Reference data (illustrative)

Spectra follow the intact-cell MALDI-TOF *Aeromonas* setting described by Surek *et al.* [1]. For each one-against-all experiment, 90% of spectra train feature extractors and the MLP; 10% are held out for blind testing (Table 1).

2.2. Architecture

The MS-ML Library implements all algorithms; the GUI orchestrates preprocessing, peak selection, feature extraction with cross-validation, and classification.

2.3. Preprocessing

2.3.1. Import and normalization

Spectra are two-column text files (m/z, intensity). Min–max normalization maps intensities to $[0, 1]$:

$$N(x, x_{\max}, x_{\min}) = \frac{x_i - x_{\min}}{x_{\max} - x_{\min}} \,. \tag{1}$$

2.3.2. Reshaping

Each spectrum is binned onto W evenly spaced grid points over the global m/z range. Let G map raw indices to grid bins via normalized coordinates [5]:

$$G(x; W, x_{\max}, x_{\min}) = \mathrm{round}\left(W \cdot N(x; x_{\max}, x_{\min})\right) \,, \tag{2}$$

with N from Eq. (1). Per-bin intensity is the maximum of contributing points; the reported m/z is the mean of their positions (see full manuscript for tie-breaking and edge bins).

2.3.3. Alignment

Optional alignment uses Icoshift [6] on user-selected m/z windows after reshaping.

2.3.4. Fuzzy moving average (FMA) peaks

At position p, a triangular-window local average $S(x; p, w_h)$ (Eq. 3) highlights local maxima as candidate peaks; w_h is half-width and T the discrete triangular weight (same functional form as ω for Watch Points below).

$$S(x; p, w_h) = \frac{\sum_{i=p-w_h}^{p+w_h} x_i \cdot T(2w_h + 1, i)}{2w_h + 1} \,. \tag{3}$$

2.4. Best peaks selection

From FMA candidates, **correlation-based** selection scores each peak's presence/absence vector against class labels (Pearson); peaks with $|\rho| > \tau$ survive. **Probability-based** selection uses class-conditional presence probabilities for discriminancy and exclusivity (see full text).

2.5. Feature extraction with cross-validation

All GA-driven extractors embed K-fold cross-validation: a configuration is retained only if the mean of training and validation correlations improves [5].

2.5.1. Watch Points

For odd window length L, triangular weights $\omega(L, n)$ are

$$\omega(L, n) = \begin{cases} \dfrac{2n}{L+1} & 1 \le n \le \dfrac{L+1}{2} \\ 2 - \dfrac{2n}{L+1} & \dfrac{L+1}{2} + 1 \le n \le L \end{cases} \,. \tag{4}$$

(Even L uses the paired formulas in the repository manuscript.) Feature W_k for watch point k is the dot product of $\omega(L_k, \cdot)$ with the binary peak vector; the GA optimizes half-widths $\{L_k\}$.

2.5.2. Feature Resonance

Peaks are encoded via a sum of M sinusoids with GA-tuned parameters:

$$R(x; \mathbf{a}, \mathbf{b}, \mathbf{c}) = \sum_{i=1}^{M} a_i \sin(b_i + c_i x) \,. \tag{5}$$

Fitness is the mean absolute Pearson correlation between R and labels; variants operate on m/z grids, best-peak intensities, m/z values, or watch-point vectors.

2.6. Classification and metrics

Features from enabled extractors are concatenated and fed to an MLP (Levenberg–Marquardt). Accuracy, precision, recall, and F1 follow standard definitions [7]. ARFF export enables external classifiers (e.g. WEKA).

3. Results (summary)

Table 1: Reference partitioning (same as full manuscript).

Species	Training (n)	Test (n)	Total (n)
A. hydrophila	511	56	567
A. caviae	511	56	567
A. veronii/sobria	286	30	316
A. trota	511	56	567

Held-out test accuracies and F1-scores on this reference task are reported in the full article; the smallest test fold (*A. veronii/sobria*, $n = 30$) warrants cautious interpretation.

4. Conclusion

MS-ML Studio implements a closed pipeline from text spectra through GA-regularized feature extractors to MLP classification. This compact note documents the methodological core; extended results, discussion, limitations, and availability match `latex/articlelsevier/main.tex`.

Limitations (pointers).. Plain-text spectra only (no native mzML/mzXML), MATLAB or MATLAB Runtime dependency, fixed MLP backend, and GA hyperparameters (population, generations, stopping rules) set via the GUI—full defaults and reproducibility checklist belong in the long manuscript or supplemental material.

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